



Projects | Praveen | March 24 2023

Orlan-10 Uncrewed Aerial Vehicle (UAV)

Orlan-10 is a medium-range, multi-purpose unmanned aerial vehicle (UAV) used by Russia’s armed forces.

Project Type

Uncrewed aerial vehicle

Operator

Russian Armed Forces

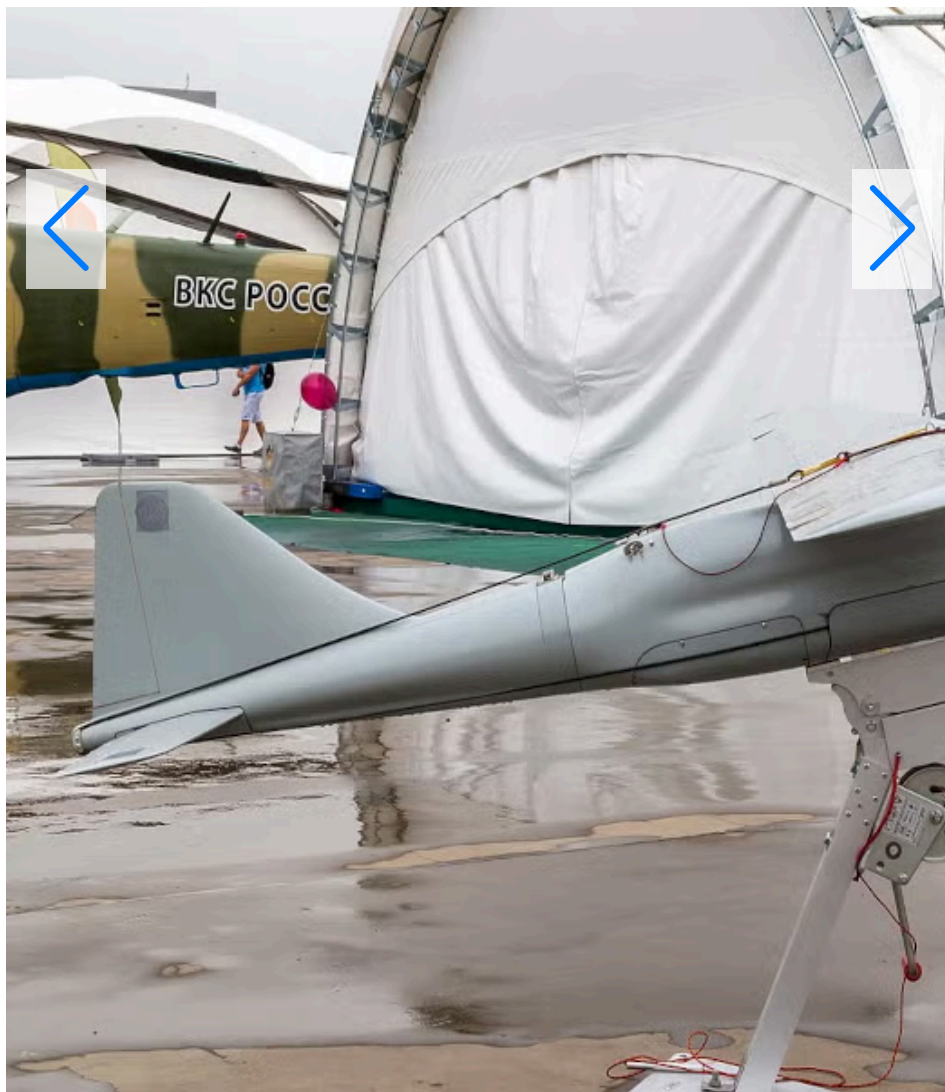
Wingspan

3.1m

Maximum Take-off Weight

16.5kg

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The Orlan-10 is a multifunctional, medium-range unmanned aeria

Wikipedia.



The Orlan-10 is a medium-range, multi-purpose uncrewed aerial vehicle (UAV) developed by Russian firm Special Technology Center LLC (ООО Специальный Технологический Центр) in St Petersburg. The UAV is in service with the Ministry of Defence of the Russian Federation.

It is intended for a variety of missions including aerial reconnaissance, observation, monitoring, search and rescue, combat training, jamming, detection of radio signals, and target tracking in hard-to-reach terrains.

The export variant, designated Orlan-10E, was exhibited at the 19th International Air and Space Fair (FIDAE) held in March 2016. Rosoboronexport announced its plans to promote the Orlan-10E UAV to foreign countries in June 2016.

Russia deployed a swarm of drones consisting of Orlan-10, Forpost, and Eleron-3 UAVs against enemy forces for the first time in Kavkaz-2020 exercises held in September 2020.

In May 2022, the Russian Ministry of Defence introduced strike versions of the Orlan-10 UAV through a video recording of a special military operation in Ukraine. The upgraded military drones were used to carry out aerial reconnaissance and precision strikes on enemy positions.

Orlan-10 drones have also been actively used by Russia in the Syrian Civil War.

Orders and deliveries of the Orlan 10s

Production on the Orlan-10 [UAV](#) commenced in 2010. The unmanned aviation unit of the Russian military base in Armenia received Orlan-10 aerial systems in October 2015, to perform aerial reconnaissance in high-mountain areas.

The Eastern Military District of the Russian Armed Forces, located in Sakhalin, Russia, took delivery of five Orlan-10 unmanned aerial vehicles in March 2016. The UAV is also operational with the Eastern Military District in Chukotka, as well as the Black Sea Fleet.

A batch of UAVs entered service with the motorised rifle unit of the Eastern Military District, stationed in the

Trans-Baikal Territory, in December 2019.

The UAV entered service with the motorised riflemen of the Southern Military District (YuVO) in the North Caucasus in July 2020. The mountain motorised rifle brigade of the Central Military District, stationed in the Republic of Tyva, took delivery of the UAV in December 2020.

In October 2020, Divisions of the mountain motorised rifle brigade of the Central Military District (CMD), based in the Republic of Tuva, received Orlan-10 UAVs under the state defence order.

Russia supplied Orlan-10E multi-purpose drone systems to Kazakhstan and Uzbekistan in November 2021. According to the Russian news agency TASS, Russia delivered more than 50 Orlan-10E drones to its customers in 2021.

Orlan-10 UAV design and features

The Orlan-10 complex consists of unmanned aerial vehicles, launch and recovery systems, ground control segment, and swappable payloads.

The UAV features modular design with high-wing configuration, while its tail section comprises a tailplane and a vertical stabiliser.

It has a wingspan of 3.1m and its aerodynamic fuselage is 2m long. The empty weight and maximum take-off weights of the UAV are 12.5kg and 16.5kg respectively.

The drone is launched using a collapsible catapult and recovered using a parachute landing system.

Payloads carried on Orlan-10 unmanned aerial vehicle

The UAV's modular design is provided with several interchangeable payloads allowing for increased mission flexibility.

It carries a day-light camera, a thermal imaging camera, a video camera and a radio transmitter in a gyro-stabilised camera pod that is fitted under the fuselage. The cameras provide real-time intelligence, 3D maps, surveillance, and aerial reconnaissance of ground-based targets.

The imagery, video and other sensor data collected by the payloads are transmitted to the ground control station in real-time, through a data link using 3G/4G cellular networks.

The Orlan-10 is fitted with electronic warfare capability and can differentiate between friendly and enemy means of transmitting of information. It can mount interference transmitters and set up zones for cellular jamming.

Command and control modes of the Orlan-10

The Orlan-10 can execute missions in both autonomous and remote-control modes. An autopilot aboard the drone enables it to operate autonomously based on pre-defined waypoints. In this mode, the flight path can be modified by the operator during the flight.

The drone can be remotely operated by its associated ground control station, installed on the MP32M1 [command and control](#) vehicle. Developed by Russian state-owned Roselectronika, the ground segment can control up to four aerial vehicles simultaneously.

In the remote-control mode, the operator at the mobile ground segment sends command and control data to the drone through the digital data link.

Orlan-10 UAV propulsion and performance

The Orlan-10 is powered by an internal combustion engine, driving a two-blade propeller mounted in the nose.

The drone can transmit telemetry data to a range between 120km and 600km from the launch station and endure for approximately 18 hours. Its cruising and maximum speeds are 110km/h and 150km/h respectively.

The UAV can fly at a maximum altitude of 5km in all weather conditions in difficult terrains. It can withstand wind speeds of 10m/s and operate in temperatures ranging from -30°C to 40°C.

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